6. KEY RESULTS

Training Loss Improves With Scale

Dataset (Scale)	Eval Loss	Trend
WikiText-2 (4k)	3.729	Improving
WikiText-2 (16k)	3.685	Improving
WikiText-2 (Full 23.7k)	3.526	Improving
WikiText-103 (64k)	**3.496**	**✓Best**

Overall Trend: Mean vs. Best-Case BLEU

OVERALL TREND: On average across all parameter combinations, iterative refinement slightly underperforms the baseline (negative mean BLEU delta). However, **best-case settings** tell a very different story, revealing massive gains that scale directly with data availability.

Dataset	Mean Δ	Best-Case Δ
4k WT2	-0.018	+0.017
16k WT2	-0.010	+0.045
64k WT103	**-0.013**	**+0.092**

BEST CONFIGURATION DISCOVERED

$N = 25$ | $\tau = 0.8$ | $r = 30\%$

Baseline BLEU: 0.2933

Relative Gain: ▲ 31%

Diffusion BLEU: 0.3853

8. CONCLUSIONS

- **Conditionally Superior:** Diffusion-style inference works, but depends critically on $\tau \times N \times r$ interaction.
- **Scale Unlocks Gains:** Shifting from 4k to 64k data increased best-case delta from +0.017 to +0.092.
- **Moderate is Best:** 30% masking is the sweet spot. It validates LLaDA's intuition for small models.
- **Metrics Deceive:** BLEU must be paired with unresolved mask counts; standard metrics hide failures.

FUTURE WORK

- **Learned Scheduling:** Adaptively anneal τ over iterations.
- **Diffusion Fine-tuning:** Train BERT with variable masking (0-50%).
- **Larger Backbones:** Transfer to BERT-large or RoBERTa.
- **Domain Text:** Apply to medical/code domains where errors are costly.